MAC ATTACK · SENIOR CAPSTONE · PROJECT PROPOSAL

MACHINE GUARDIAN

An adaptive INT4/INT8 FPGA TPU for acoustic machine-condition monitoring

PROPOSED SYSTEM · OCTOBER 2026

Platform: Avnet ZedBoard (Zynq-7020) **Team:** MAC Attack **Timeline:** October 2026–April 2027

Executive Summary

Machine Guardian will use a custom FPGA-based AI accelerator to detect abnormal operating sounds from machines such as fans and motors. Rather than running only a conventional software classifier, the team will implement neural-network arithmetic using its own folded systolic array on the ZedBoard.

The distinctive experiment is **adaptive precision**: process a feature window first with low-cost INT4 arithmetic, then rerun in INT8 when the result is close to an anomaly decision boundary. The research will determine whether this reduces average energy per inference while preserving detection quality. These savings are a **hypothesis to test**, not an established result.

CENTRAL RESEARCH QUESTION

Can an adaptive INT4/INT8 systolic-array TPU reduce the energy cost of acoustic anomaly detection without an unacceptable loss in accuracy?

Why It Matters

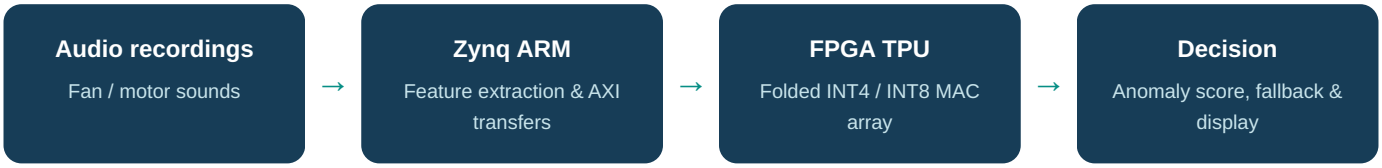
Real-world application

Acoustic monitoring can flag unusual machinery sounds without intrusive installation. It may support maintenance decisions, but this prototype will detect anomalies—not predict exact failure dates.

Hardware research

INT4, INT8, folding, PE count, local memory, and clock rate create a measurable design space involving accuracy, computation time, silicon resources, and energy.

Proposed System



The neural network will be **trained offline on a computer**. Quantized weights and features will be transferred to the Zynq system. The ARM cores manage preprocessing and control; the programmable logic accelerates the network's arithmetic. AXI—not slow UART input—is the intended operational interface.

Research Design & Experiments

Operating mode	Meaning	Research purpose
Fixed INT8	All eligible neural-network layers execute using 8-bit precision.	Accuracy and energy reference.
Fixed INT4	All eligible layers execute in 4-bit precision.	Minimum-precision baseline and accuracy loss.
Adaptive INT4 → INT8	Start INT4; rerun uncertain windows in INT8 using a calibrated threshold band.	Test whether selective higher precision delivers a useful energy/accuracy tradeoff.

Primary dataset: DCASE 2020 Task 2, initially the fan subset, to train/evaluate acoustic anomaly detection. **Secondary comparison:** the team's existing MLPerf Tiny anomaly-detection autoencoder workload. The team should choose one primary test split and lock its evaluation procedure before benchmarking.

ML measurements

- ROC AUC and partial AUC
- False-positive/false-negative behavior
- INT4 and INT8 anomaly-score agreement
- Adaptive rerun fraction

Hardware measurements

- Cycle count and inference latency
- LUT, FF, BRAM and DSP utilization
- Achievable clock / timing slack
- Power and energy per inference
- Data movement and PE utilization

Initial Technical Targets (not verified performance)

Dimension	Initial target	Decision rule
Board	ZedBoard XC7Z020	Existing board; no replacement FPGA purchase.
Array	4×4 bring-up; 8×8 baseline; 16×16 stretch	Scale only after resource and timing reports.
Arithmetic	Signed INT8 & INT4; accumulation width validated against model	Bit-exact checks against Python.
Clock	80 MHz PL clock starting target	Generate real PL clock; verify timing closure.
Accuracy	Adaptive AUC loss ≤0.03 vs. fixed INT8	Measure on held-out dataset.

Dimension	Initial target	Decision rule
Energy	Stretch: ≥20% lower average energy vs. fixed INT8	Include reruns and data movement; measure consistently.
Neural model	Compact fully connected autoencoder	Adjust size after software quantization results.

Important design issue: INT4 values placed inside an always-active INT8 datapath do not automatically save energy. Demonstrating operand isolation or low-precision arithmetic efficiency is part of the hardware work.

Implementation Roadmap

Stage	Activities	Evidence / deliverable
1 · ZedBoard bring-up Oct 2026	Verify board, power and programming; create Zynq project; generate PL clock; test AXI registers.	Working hardware bitstream and ARM-to-PL test.
2 · TPU port Oct–Nov	Port existing SystemVerilog systolic array; begin 4×4 or 8×8 INT8; implement buffer/control interface.	FPGA matrix outputs match Python bit-for-bit.
3 · ML baseline Oct–Dec	Train normal-sound autoencoder; establish held-out AUC; quantize and export weights.	FP32, INT8, INT4 software baselines.
4 · End-to-end inference Dec–Jan	Integrate layers, activation/requantization, transfers and anomaly score.	INT8 anomaly inference running on ZedBoard.
5 · Adaptive controller Jan–Feb	Integrate INT4 execution and decision-boundary fallback; validate correctness and overhead.	Three precision modes measured consistently.
6 · Evaluation and demo Feb–Apr 2027	Benchmark accuracy, timing, area and energy; demonstrate recorded audio then optional live fan; relate results to ASIC/OpenLane exploration.	Capstone demo, plots and technical report.

Demonstration Plan

First demo: A PC provides numerical features extracted from known fan recordings. The Zynq ARM loads them via AXI into the accelerator. The FPGA produces outputs, and software calculates/displays an anomaly score along with measured cycles.

Final demo: The audience selects normal and abnormal machine recordings and watches the system's prediction, anomaly score, and INT4/INT8 mode. If time and hardware integration permit, use the ZedBoard's onboard audio codec to capture a real low-voltage desk fan. A real fan may require separate calibration or retraining; dataset accuracy must not be presented as live-fan accuracy.

Scope & Risks

In scope

- Quantized inference and FPGA implementation
- AXI-linked Zynq/PL operation
- Fixed vs. adaptive precision comparison
- Hardware benchmarks and ASIC-relevant insights

Not promised

- Predicting exact failure date
- Beating a high-end GPU in throughput
- Fully autonomous industrial deployment
- Guaranteed energy savings

CRITICAL PROJECT SAFEGUARD

The core success criterion is **correct INT8 anomaly inference on the actual ZedBoard**. Adaptive precision, energy savings, and live sensing are successive improvements—not prerequisites for proving the accelerator works.

What We Need From the Professor

Confirm the research emphasis: **adaptive precision as the novelty**, acoustic anomaly detection as the practical demonstration, and resource/energy/accuracy measurements as the scientific evidence. Request access to a ZedBoard, basic programming accessories and laboratory power-measurement instruments.

Project brief prepared for discussion and planning. Baseline RTL/testing results are recorded in the team's existing MAC Attack project log; the Machine Guardian application and targets in this document are proposed work, not completed implementation. Dataset: dcase.community/challenge2020/task-unsupervised-detection-of-anomalous-sounds